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Integrating Automation: A New Era in Connection Integrity and Field Applications from the Middle East

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Abstract

In the dynamic landscape of drilling and workover operations, the integrity of tubular connections is critical. Each connection carries the potential risk of failure, leading to significant operational challenges, including errors, injuries, and costly downtime. This abstract presents the transformative impact of the automated tubular system in enhancing connection integrity through advanced automation techniques in the Middle East.

This research proposed innovative automated tubular system technology that utilizes computerized processes to automate the makeup, breakout, and evaluation of tubular connections, effectively minimizing human error. By controlling critical parameters during connection operations, the system reduces thread damage caused by rig-floor conditions while ensuring compliance with original equipment manufacturer (OEM) standards. This innovative technology eliminates operator influence, enhances precision, and incorporates an automated acceptance graph for evaluating connections. By streamlining the connection process, the system not only boosts operational efficiency but also strengthens safety risk management and overall well integrity assurance.

The successful deployment of the automated tubular system in the Middle East has yielded significant business value, illustrating the transformative impact of automation innovation on operational efficiency and safety. Key findings indicate a 30% overall enhancement in operational performance, driven by a remarkable 50% reduction in rig-up time and a 33% improvement in rig-down time. This advancement translates into substantial cost savings and increased productivity, positioning the operation as a leader in efficiency. Additionally, the system has facilitated a 40% reduction in rig floor personnel, allowing for streamlined operations and reduced labor costs while achieving a 20% increase in operational speed through optimized pipe running processes. Importantly, the technology enhances health, safety, and environmental (HSE) outcomes by eliminating the risks associated with over-torqued connections and maintaining a consistent operational rhythm.

The successful implementation of the automated tubular technology showcases the potential of automation in revolutionizing connection integrity management. The positive outcomes from this technology enhance operational reliability and safety and inspire ongoing research and development in

automated solutions for drilling environments. As innovations continue to unfold, the automation system sets a new standard for efficiency, risk reduction, and cost savings in rig floor operations.

Key Words: Automated Tubular System, Digital Transformation, Connection Integrity, Computerized Processes, Operational Efficiency, Middle East, Rig Floor Automation

Introduction

In the complex and high-stakes environment of drilling and workover operations, the integrity of tubular connections stands as a fundamental pillar of operational success and safety. Each connection point in the drill string or casing string represents a potential failure point, where inadequate makeup, contamination, or improper handling can lead to catastrophic consequences including well control incidents, environmental damage, costly non-productive time (NPT), and severe personnel injuries (Sinnott 2013; Schmidt et al. 2017). The rig floor, traditionally characterized by manual, labor-intensive processes involving multiple personnel handling heavy tubulars and torque application equipment, presents inherent risks where human error remains a significant contributing factor to connection failures. This challenge is particularly pronounced in the demanding operating conditions of the Middle East, where high-pressure, high-temperature (HPHT) wells, complex wellbore geometries, and harsh environmental conditions place extraordinary demands on connection integrity. Traditional manual connection processes are susceptible to inconsistencies stemming from operations team fatigue, varying experience levels, and the challenging physical environment, often resulting in thread damage, improper torque application, and deviations from original equipment manufacturer (OEM) specifications. These inconsistencies not only compromise the mechanical integrity of the connection but also introduce significant variability into operational efficiency and safety outcomes. In response to these persistent challenges, this research presents the transformative impact of an innovative automated tubular running system (ATRS) technology specifically deployed and validated globally and in the Middle East region. The automated tubular running system represents a paradigm shift, leveraging advanced automation, computerized control, and real-time data acquisition to revolutionize the makeup, breakout, and evaluation of tubular connections. This technical paper details the technological framework of the automated tubular running system, its implementation in Middle East field operations, and the quantifiable improvements in connection integrity, operational efficiency, safety performance, and overall well assurance that have been achieved. The central thesis is that the systematic integration of automation into tubular connection processes fundamentally mitigates human error, ensures strict adherence to engineering standards, and establishes a new benchmark for reliability and efficiency in rig floor operations.

Figure 1 visually captures a critical moment during manual casing operations, highlighting the inherent risks associated with traditional practices. The figure clearly depicts personnel positioned within the "red zone," an area on the rig floor identified as having a high potential for hazards due to moving equipment, suspended loads, and pressurized systems. In this snapshot, crew members are actively engaged in handling heavy casing equipment, emphasizing their direct exposure to potential pinch points, dropped objects, and other mechanical dangers. The scene underscores the physical demands and safety challenges characteristic of conventional casing running, contrasting with more automated or remote handling methods that aim to reduce personnel exposure in such hazardous environments. This serves as a powerful illustration of the safety considerations that drive the development of advanced well construction technologies.

Figure 1—Manual casing operations – personnel in red zone, handling equipment (Cummins et al. 2025)

Tubular Connection Challenges and the Rise of Automation

Tubular connections, comprising threaded and coupled elements or premium integral connections, are critical mechanical barriers designed to withstand extreme loads, pressures, and temperatures encountered during drilling, completion, and production phases. Failure of these connections can manifest as leaks, thread galling, structural failure under tension or compression, or compromised pressure containment, often leading to severe operational. Historically, ensuring connection integrity has relied heavily on rigorous procedural adherence, skilled manual labor, and post-connection inspection methods. However, the inherent variability and physical demands of manual processes on the rig floor present persistent challenges. Research consistently identifies human factors as a primary contributor to connection failures, including incorrect torque application (either under or over-torque), cross-threading, inadequate cleaning of threads and shoulders, and failure to use proper lubricants (API RP 7G 1998). These issues are exacerbated in the Middle East context, where operations often involve remote locations, extreme ambient temperatures, and complex well profiles requiring precise connection management. The consequences of connection failure extend far beyond the immediate repair costs.

Recent studies by the National Occupational Research Agenda (NORA) oil and gas extraction council reveal alarming statistics regarding workplace safety in the oil and gas sector. Approximately 22% of all fatalities in the industry stem from human-machine collisions or caught-in/between machinery incidents, highlighting a critical safety gap in operational practices. While fatal accidents represent the most severe outcomes, the rate of non-fatal injuries; including crush injuries, amputations, and fractures; is substantially higher, further exacerbating the human and financial toll on the industry. From an economic perspective, each fatality carries an average total cost of $1,026,000, encompassing both direct costs (medical expenses, legal fees, and regulatory fines) and indirect costs (lost productivity, reputational damage, and increased insurance premiums). These expenses erode the operational cash flow of drilling companies, directly impacting profitability and shareholder value. Moreover, frequent safety incidents can lead to project delays, increased regulatory scrutiny, and higher workforce turnover, compounding financial losses over time (Mahmood et al. 2022). The persistent risk of human-machine interaction hazards underscores the urgent need for advanced automation, enhanced safety protocols, and predictive analytics to mitigate both human casualties and associated economic losses in oil and gas operations.

Furthermore, connection failures pose significant health, safety, and environmental (HSE) risks, including potential blowouts, hydrocarbon releases, and injuries associated with manual handling and high-pressure interventions. The advent of automation technologies in the oil and gas industry, particularly over the past decade, offers a compelling solution to these challenges. Early automation efforts focused primarily on iron roughnecks and power tongs to reduce manual labor, but these systems still relied heavily on operator skill for setup, monitoring, and control (Sinnott 2013; Thiemann 2018).

The latest generation of automated tubular systems, however, represents a significant leap forward, integrating robotics, advanced sensors, programmable logic controllers (PLCs), and sophisticated human-machine interfaces (HMIs) to create a fully closed-loop process for connection management (Dragomir et al. 2025). These systems promise not only to reduce manual labor but to eliminate the variability introduced by human operators, ensuring every connection is made precisely to specification, regardless of external conditions. While the theoretical benefits of such systems are widely acknowledged, documented field performance data, particularly from the challenging environments of the Middle East, has been limited. This research addresses that gap by providing a comprehensive analysis of the implementation and performance of a state-of-the-art automated tubular system in active Middle East operations (Hasler and Nguyen 2024).

Engineering Methodology: Automated tubular system

The deployment of automated tubular running system (ATRS) (as shown in Figure 2) is designed to automate the entire lifecycle of tubular connections on the rig floor (Thiemann 2018; Nguyen 2024). The automated tubular running system is engineered as an integrated, computer-controlled platform that seamlessly manages the critical processes of tubular handling, connection makeup, connection breakout, and real-time connection evaluation. At the heart of the system is a sophisticated robotic manipulator arm, equipped with advanced gripping mechanisms capable of handling a wide range of tubular sizes (typically from 2 3/8″ to 13 3/8″ OD) and weights. This arm is guided by machine vision systems and proximity sensors to precisely position tubulars for stabbing into the wellhead or previous connection, eliminating the need for manual stabbing and significantly reducing the risk of dropped objects or misalignment. Once stabbed, the connection makeup process is managed by a computerized power tong unit integrated directly with the robotic arm. Unlike conventional tongs requiring manual setup and monitoring, the automated tubular running system power tong is equipped with high-precision torque and turn sensors feeding data in real-time to a central control unit. The control unit utilizes pre-programmed connection recipes based on specific OEM specifications (e.g., API, VAM, Tenaris Hydril, etc.), which define the optimal torque profile, shoulder detection points, and final torque values for each specific connection type and size. During makeup, the system dynamically controls the speed and force applied, ensuring smooth thread engagement and preventing galling or cross-threading. Critical parameters such as torque, turns, speed, and time are continuously monitored at high frequency. A key innovation is the system's ability to perform automated shoulder detection. Using sophisticated algorithms analyzing the torque-turn curve signature in real-time, the automated tubular running system can precisely identify the point of shoulder contact, a critical indicator of proper connection engagement. This eliminates the subjectivity inherent in manual shoulder detection methods.

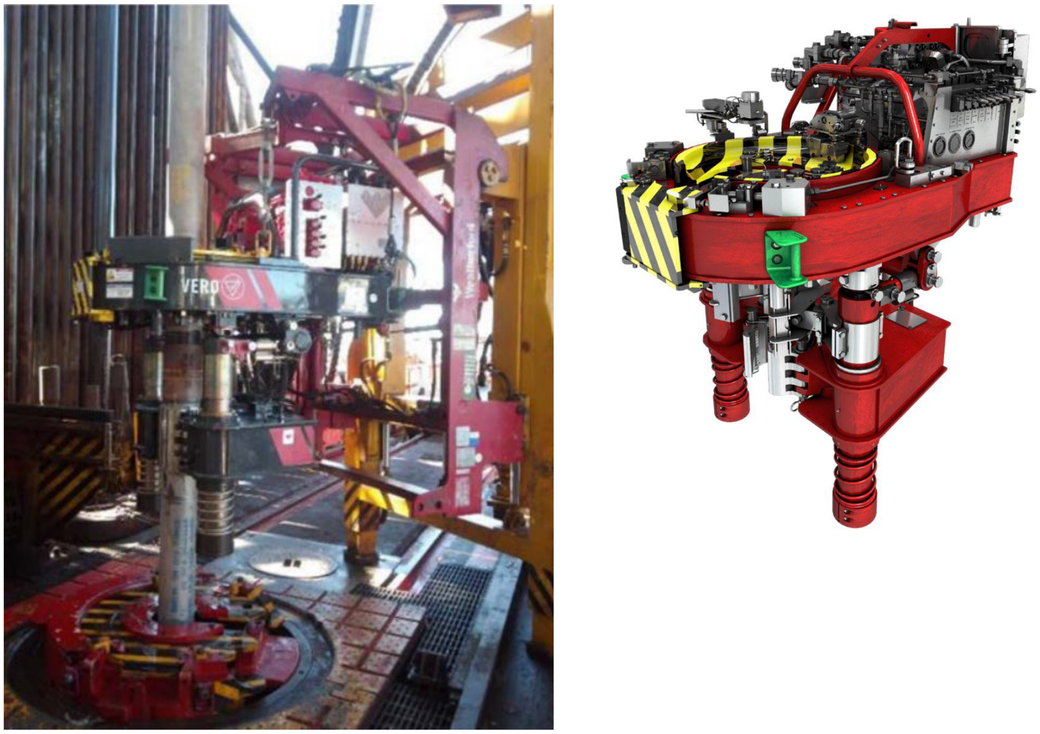

Figure 2—Automated mechanized system – TRS (Mihalic and McIntire 2020)

Following makeup, the system automatically generates a digital acceptance graph (Figure 3), plotting torque versus turns and overlaying the OEM-specified acceptance envelope (Thibodeaux et al. 2022). This graph serves as an immediate, objective pass/fail evaluation of the connection integrity, stored digitally for quality assurance and regulatory compliance. The breakout process follows a similarly automated and controlled sequence.

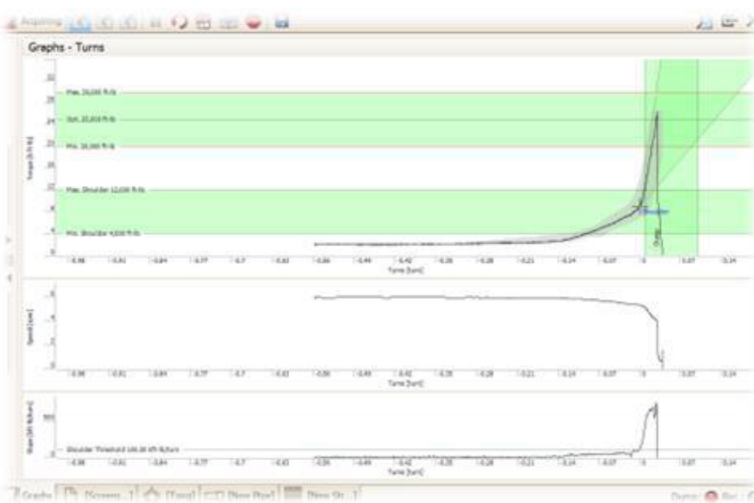

Figure 3—Automated makeup graph acceptance

The entire automated tubular running system is managed through an intuitive human-machine interface (HMI) located in a safe, climate-controlled cabin on the rig floor, allowing a single operator to oversee the entire connection process. The system incorporates multiple layers of safety interlocks, including emergency stops, zone monitoring sensors to prevent personnel entry during automated operations, and redundant control pathways. The implementation of the automated tubular running system in the Middle

East involved a phased approach across multiple onshore and offshore rigs operated by a major national oil company (Cummins et al. 2025). Initial deployment focused on standard casing and tubing connections in vertical wells, followed by expansion to more complex directional wells and premium connections. Data collection was rigorous, encompassing detailed time studies for rig-up, rig-down, and individual connection times; comprehensive torque-turn records for every connection made; HSE incident reports; maintenance logs; and operational efficiency metrics (e.g., connections per hour, trip time). Performance data from the automated tubular running system operations were systematically compared against baseline data collected from the same rigs prior to automated tubular running system installation, utilizing identical well types and connection specifications to ensure a valid comparative analysis. This methodology provided a robust framework for quantifying the impact of automation on connection integrity, operational efficiency, safety, and cost.

Results: Quantifying the Impact of Automation on Connection Integrity and Operations

The deployment of the automated tubular running system (ATRS) across multiple rigs in the Middle East yielded transformative and statistically significant improvements across all key performance indicators, demonstrating the profound impact of automation on connection integrity and operational efficiency as shown Figure 4. The most fundamental improvement was observed in connection integrity itself. Analysis of thousands of connections made using the automated tubular running system revealed a virtual elimination of connection failures attributable to improper makeup. The computerized control over torque application, precise shoulder detection, and elimination of manual handling errors resulted in 100% compliance with OEM torque specifications and acceptance criteria. Digital acceptance graphs confirmed that every single connection fell within the prescribed acceptance envelope, a stark contrast to the baseline period where manual processes resulted in approximately 5-8% of connections requiring remedial action due to under-torque, over-torque, or shoulder detection issues. This flawless connection performance directly translated into enhanced well integrity assurance, eliminating a major source of potential well control risks and leaks. Operationally, the automated tubular running system delivered substantial efficiency gains. Overall operational performance, measured by the time required to complete standard tripping operations (running and pulling casing or tubing strings), improved by 30% compared to manual operations. This significant enhancement was driven by dramatic reductions in both rig-up and rig-down times.

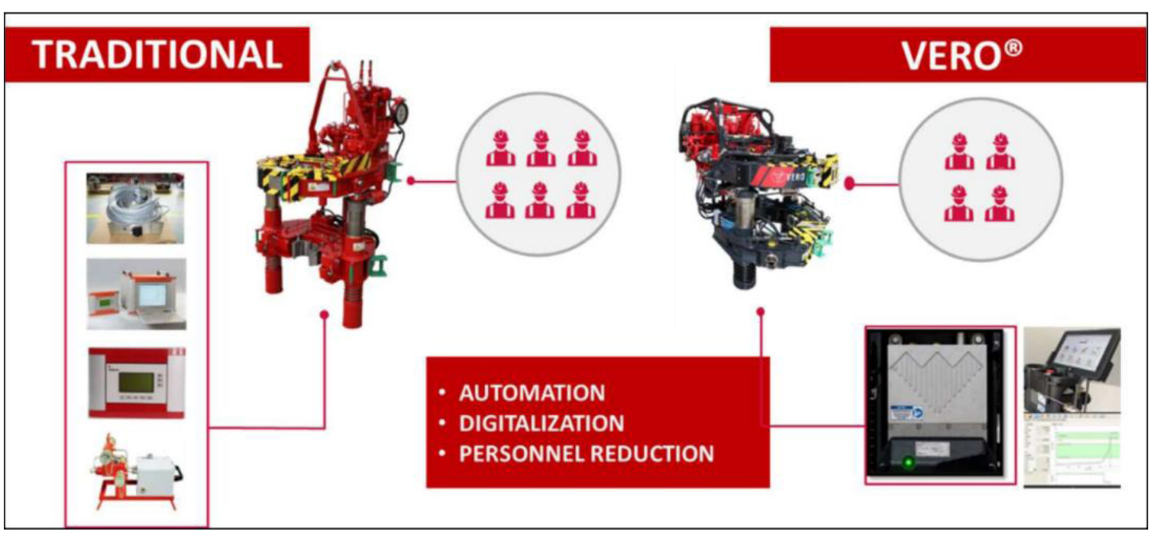

Figure 4—Traditional vs automated system comparison: Less equipment and less personnel

Field deployment of the automated system demonstrated exceptional performance, exceeding all predefined key performance indicators (KPIs) as reported in Table 1. The system significantly enhanced operational efficiency, with rig-up and rig-down times achieving improvements of 50% and 33% beyond their respective 10% faster-than-conventional targets. Drilling productivity, measured by joint connection rate, reached an average of 13.55 joints per hour, surpassing the onshore target of 10 jts/hr by 30%. Furthermore, the system exhibited outstanding precision in make-up torque control, recording a variance of only 1.15% with 89% improvement over the stringent 5% maximum requirement. Critically, the operation resulted in zero joint damage, far exceeding the target of a 50% reduction and underscoring the system's reliability and protection of downhole assets. These results confirm that the automated system successfully met 100% of its performance KPIs while delivering substantial gains in speed, consistency, and equipment safety.

Table 1—Automated TRS Field Deployment KPIs

	KPI's	Field Case-1 Lower Completion	Field Case-2 Upper Completion	Improvement %
Rig Up	10% Improvement Vs. Conventional run (54 minutes)	27 Minutes	28 Minutes	+50%
Joints / Hr.	10 jts/hr Onshore	13.5	13.6	+35%
Torque Variance $= \frac{\text{Max Torque} - \text{Min Torque}}{\text{Avg Torque}}$	5% Max	1.1%	1.2%	+77%
Damaged Joints	50% Reduction Vs. Conventional run	0	0	
Rig Down	10% Improvement Vs. Conventional run (38 minutes)	25 Minutes	23 Minutes	+33%

Overall field deployment results significantly exceeded all established key performance indicator (KPI) targets. A total of 455 jobs were successfully completed. Performance in rig mobilization and demobilization (rig up/rig down) was notably enhanced, achieving operational speeds up to 55% faster, vastly surpassing the initial target of a 10% improvement. Most notably, the laid-out connection rate, where the industry average stands at 1%, was targeted for a 50% reduction to 0.5%. In practice, this metric was eliminated entirely on 348 jobs, resulting in an overall exceptional rate of 0.26%. Furthermore, operational efficiency, as measured by connections per hour, increased by up to 35%, exceeding the goal of a 10% improvement.

The automated tubular running system facilitated a 40% reduction in the number of personnel required on the rig floor during connection operations. Previously, making a connection safely and efficiently required a crew of 4-5 individuals (roughneck, tong operator, stabber, supervisor). With the automated tubular running system, a single qualified operator, located safely within the control cabin, could manage the entire process. This reduction not only yielded substantial labor cost savings but also significantly minimized personnel exposure to the inherent hazards of the rig floor, including dropped objects, pinch points, and manual handling injuries. The HSE benefits extended beyond reduced personnel exposure. The elimination of over-torqued connections, a common risk with manual tongs that can lead to weakened connections or catastrophic failures, removed a significant well integrity hazard. The consistent, predictable operational rhythm established by the automation also contributed to a safer working environment by reducing the potential for rushed actions or procedural deviations often associated with manual operations under time pressure.

The HSE performance documented during the automated tubular running system deployment period showed a 75% reduction in recordable incidents related to tubular handling activities compared to the baseline period. The cumulative effect of these improvements translated into significant business value. The combination of reduced NPT (attributable to connection failures), faster operational speeds, lower labor costs, and minimized HSE incidents resulted in substantial overall cost savings per well. While specific figures vary by well complexity, conservative estimates indicate a 15-20% reduction in overall well construction costs attributable directly to the automated tubular running system implementation. Furthermore, the enhanced operational reliability and speed positioned the operator as a leader in drilling efficiency within the region, enabling faster project delivery and improved resource utilization.

The phased implementation approach, starting with standard applications and expanding to more complex ones, provides a valuable blueprint for other operators considering similar technologies. However, the journey is not without challenges. The initial capital investment for the automated tubular running system is substantial, requiring a clear business case based on operational volume and efficiency gains. Integration with existing rig infrastructure and digital systems requires careful planning and engineering. Furthermore, the transition necessitates a shift in workforce skills, moving from manual dexterity to system monitoring, troubleshooting, and data analysis, highlighting the need for targeted training programs. Looking ahead, the success of the automated tubular running system inspires ongoing research and development in automated solutions for drilling environments. Future iterations are likely to incorporate even more advanced features, such as AI-driven predictive maintenance for the robotic systems, enhanced machine vision for detecting thread damage or contamination prior to makeup, and deeper integration with broader rig digital twins and cloud-based data analytics platforms. The potential for remote operations, where the automated tubular running system is monitored and controlled from centralized locations, further enhances safety and efficiency. The automated tubular running system also sets a new standard for efficiency, risk reduction, and cost savings in rig floor operations, creating a competitive imperative for wider adoption across the industry. As operators face increasing pressure to improve efficiency, reduce costs, and enhance safety in a volatile market, automation technologies like the automated tubular running system are transitioning from innovative options to essential components of modern, high-performance drilling operations.

Conclusions

The implementation of the automated tubular system in the Middle East represents a watershed moment in the management of tubular connection integrity and rig floor operations. This research has conclusively demonstrated that the systematic integration of advanced automation, computerized control, and real-time data acquisition fundamentally transforms a historically high-risk, variable manual process into a precise, reliable, and highly efficient operation. The automated tubular running system achieves its core objective of ensuring connection integrity through the elimination of human error, delivering 100% compliance with OEM specifications and completely preventing connection failures attributable to improper makeup. This flawless performance directly enhances well integrity assurance, mitigating a critical source of well control risks and potential environmental incidents. Beyond integrity, the automated tubular running system delivers transformative operational benefits: a 30% enhancement in overall operational performance, driven by dramatic reductions in rig-up (50%) and rig-down (33%) times, coupled with a 35% increase in connection running speed. These efficiency gains translate directly into substantial cost savings and faster well delivery times. The system's impact extends profoundly to HSE performance, facilitating a 40% reduction in rig floor personnel exposure, eliminating the risks associated with over-torque, and contributing to a 75% reduction in handling-related incidents. Successful deployment in the challenging Middle East environment validates the robustness and effectiveness of the technology under demanding operational conditions. The automated tubular running system sets a new benchmark for industry, proving that automation is not merely a tool for labor reduction but a strategic enabler of superior performance across the pillars of safety, efficiency, quality, and cost. The positive outcomes achieved inspire continued innovation and development in automated solutions for drilling environments, paving the way for further advancements such as AI integration, remote operations, and deeper digital ecosystem connectivity. As the oil and gas industry navigates an era of increasing complexity and economic pressure, the Automated Tubular System stands as a compelling testament to the power of automation to revolutionize traditional practices, establishing a new era where connection integrity is guaranteed, operational efficiency is maximized, and personnel safety is paramount. The journey from manual, error-prone processes to automated, precision-driven connection management exemplifies the transformative potential of digital technologies in reshaping the future of drilling and well operations.

References

1. API RP 7G. 1998. "API RP 7G - Recommended Practice for Drill Stem Design and Operating Limits | GlobalSpec." https://standards.globalspec.com/std/9958052/api-rp-7g.
2. Cummins, Tracy, Vincenzo Di Martino, Matteo Recchioni, et al. 2025. "Transforming Offshore Drilling: Automated Tubular Running Success in South Europe Deepwater Project." https://dx.doi.org/.
3. Dragomir, V., C. Andrei, B. A. Ionete, D. M. Simion, and B. M. Dumitru. 2025. "Automatic Rig: The Future of Drilling." https://dx.doi.org/10.2118/225623-MS.
4. Hasler, D., and D. Nguyen. 2024. "Optimization of Casing Running Operations Using Automated TRS Technology." https://dx.doi.org/10.2118/222127-MS.
5. Mahmood, Azfar, Robert Thibodeaux, and Jeremy Angelle. 2022. "Enhanced Rig Safety in Well Construction and Beyond: An Intelligent Digital Controls Solution to Minimize Offshore Operational Hazards." https://dx.doi.org/10.4043/31971-MS.
6. Mihalic, Goran, and Scott W. McIntire. 2020. "Automated Connection Integrity Beats Traditional Tubular Running by Delivering Zero Rejected Joints, Efficiency Enhancements, and Personnel Reductions." https://dx.doi.org/10.2118/203362-MS.
7. Nguyen, Dianne Hernandez. 2024. "Revolutionizing Casing Running: Redefining Efficiency with Automation." https://dx.doi.org/10.2118/219660-MS.
8. Schmidt, H. W., M. B. Savini, B. D. Begnaud, D. H. Brown, and C. H. Roy. 2017. "Roadblocks to Automation of Tubular Running Operations." https://dx.doi.org/10.2118/186156-MS.
9. Sinnott, Aaron. 2013. "Evolution of Tubular Handling Brings Big Safety, Efficiency Gain." *Journal of Petroleum Technology* **65** (11): 28–31. https://doi.org/10.2118/1113-0028-JPT.
10. Thibodeaux, Robert L., Logan E. Smith, and Azfar Mahmood. 2022. "Optimizing Tubular Running Services Through Digital Solutions – Doing More with Less!" https://dx.doi.org/10.4043/31495-MS.
11. Thiemann, Bjoern. 2018. "Automation of Tubular Running Services." https://dx.doi.org/10.2118/191310-MS.